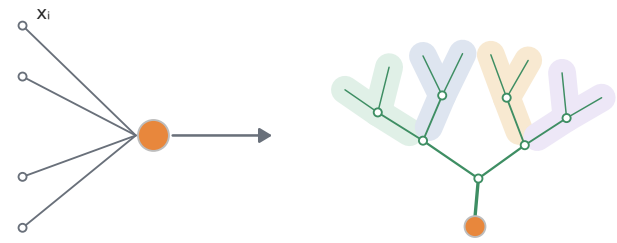
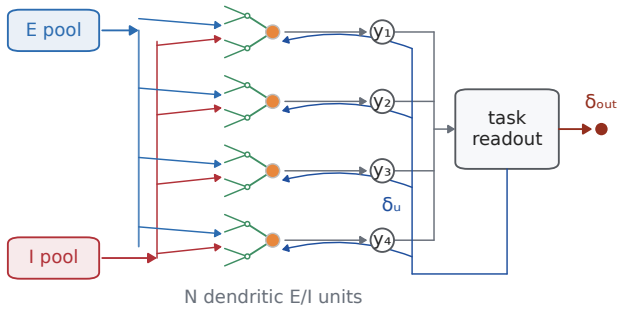
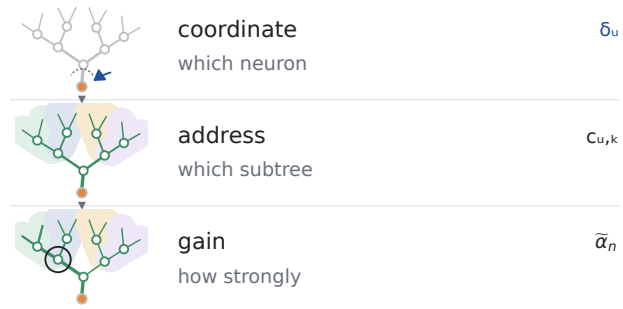


APoint unit $\rightarrow$ dendritic treeone neuronal coordinate δ_u subtree coefficients $C_{u,k}$ **B**

Network layer



N dendritic E/I units

CCoordinate $\rightarrow$ address $\rightarrow$ gain

coordinate

which neuron

 δ_u

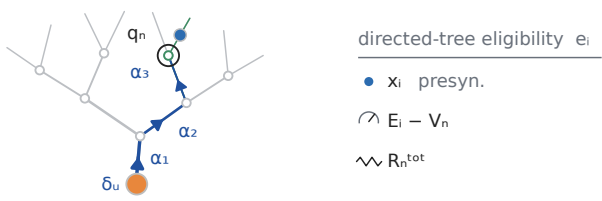
address

which subtree

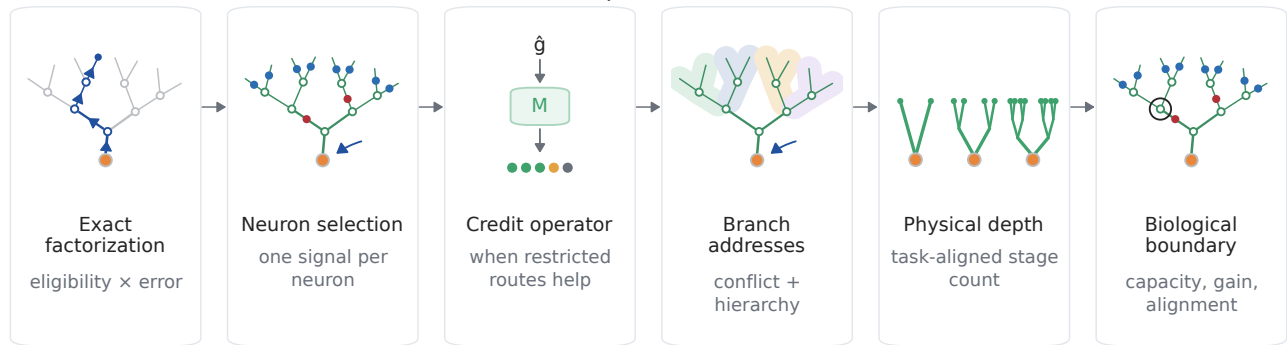
 $C_{u,k}$

gain

how strongly

 $\tilde{\alpha}_n$ **D**Local eligibility $\times$ transported errordirected-tree eligibility e_i $\bullet$ x_i presyn. $\bigcirc$ $E_i - V_n$ $\rightsquigarrow$ R_n^{tot} directed tree: $\partial \mathcal{L} / \partial g_i = [x_i R_n^{tot} (E_i - V_n)] [\partial \mathcal{L} / \partial V_n]$ general adjoint: $\partial \mathcal{L} / \partial g_i = x_i (E_i - V_n) q_n$, $q_n = R_n^{tot} \partial \mathcal{L} / \partial V_n$ **E**

Roadmap of the Results

Exact factorization
eligibility $\times$ errorNeuron selection
one signal per neuronCredit operator
when restricted routes helpBranch addresses
conflict + hierarchyPhysical depth
task-aligned stage countBiological boundary
capacity, gain, alignment